TECHNICAL ASSESSMENT

Automated Weld Defect Classification System — Design Brief

For the role of Computer Vision Engineer (Robotics & Automation)

ISSUED BY	zPROx.AI — AI & Robotics Systems Group
ROLE	Computer Vision / Machine Learning Engineer (Robotics & Automation)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	16 May 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Computer Vision Engineer role within the Robotics & Automation Systems group, where engineering teams design and deploy real-time vision systems for industrial inspection, automated quality assurance, and robotic guidance on the factory floor. This assessment presents a problem representative of the work the team undertakes, and is intended to evaluate the candidate's ability to design, implement, and clearly communicate a working solution that meets production constraints.

2. PROBLEM STATEMENT

Design and implement a system that, given an image of a weld captured from an inline camera mounted above a conveyor belt, automatically classifies the weld as either acceptable (“good”) or defective, in real time, so that downstream robotic actuators may act on the decision (for example, by diverting defective parts to a reject bin).

3. OBJECTIVE

Given a labelled dataset containing images of good welds and defective welds, the candidate shall develop a classification model and surrounding inference pipeline that:

(a) is trained on the provided dataset, and any additional publicly available data that the candidate considers appropriate;

(b) accepts an arbitrary weld image as input at inference time; and

(c) emits a clear, deterministic classification decision and associated confidence score, suitable for triggering downstream actions on a conveyor-belt sorting system.

4. FUNCTIONAL REQUIREMENTS

4.1 Input Image Support

The system shall correctly accept and process images captured under typical industrial-camera conditions, including both colour (three-channel RGB) and grayscale (single-channel) inputs, varying resolutions, and minor variations in lighting, viewing angle, and stand-off distance.

4.2 Classification Pipeline

The system shall accept an arbitrary input image, perform all required pre-processing, model inference, and post-processing internally, and return a single classification decision (at minimum: “good” or “defective”) together with a numerical confidence score, without requiring further user intervention.

4.3 Real-Time Performance

The system shall be capable of operating within the latency budget of a moving conveyor line. As an indicative target, end-to-end inference latency on a single image should not exceed 200 milliseconds on commodity hardware (CPU or a single mid-range GPU). Candidates shall report the measured latency on their chosen reference hardware.

4.4 Robustness

The system shall remain reliable under realistic variations in input quality, including changes in illumination, modest blur or focus variation, partial occlusion (e.g. residual spatter, sparks, or surface staining), and the presence of background clutter.

4.5 Generalisation

The system shall not be overfit to any single batch, weld geometry, or imaging setup. It is expected to perform reasonably on previously unseen welds drawn from related distributions. Where the candidate identifies generalisation limits, these shall be stated explicitly.

5. INPUT AND OUTPUT SPECIFICATION

Input	A single weld image in any common raster format (e.g. JPEG, PNG), captured from an overhead or oblique-view inline camera positioned above the conveyor belt.
Output	A structured result, emitted in a machine-readable format (e.g. JSON), comprising at minimum: (a) a class label ("good" or "defective"); (b) a confidence score in the range [0, 1]; and (c) optionally, a defect-type sub-label (e.g. porosity, undercut, crack, spatter, incomplete fusion) where the candidate's model supports it.

6. EVALUATION CRITERIA

Submissions will be assessed against the following criteria. Candidates are advised to address each criterion explicitly in their submission.

Criterion	Description
6.1 Classification Accuracy	Overall accuracy, precision, recall, and F1 score on a held-out test set. Particular weight will be given to recall on the "defective" class, since missed defects are costly on a production line.
6.2 Inference Latency	Measured end-to-end latency per image on the candidate's reference hardware, reported alongside the hardware specification.
6.3 Robustness	The system's behaviour under noisy, low-quality, or otherwise non-ideal inputs representative of factory-floor conditions.
6.4 Generalisation	Performance on inputs not seen during development or training, including any variation in weld geometry or imaging conditions.
6.5 Engineering Soundness	Quality of the inference pipeline as a whole — data handling, model selection, training procedure, evaluation methodology, and readiness for deployment in an industrial setting.
6.6 Clarity of Approach	The quality, structure, and correctness of the candidate's explanation of their methodology.

7. SUBMISSION REQUIREMENTS

Candidates are required to submit:

(a) the working system, including training and inference code, any saved model weights, and clear instructions to reproduce the results;

(b) a short written report describing the dataset used, model architecture, training procedure, evaluation results, and any known limitations; and

(c) a short explanation video. The video is a mandatory component of this assessment and shall cover, at a minimum:

(i) the candidate's overall approach to the problem and the rationale for the chosen technique;

(ii) a clear description of how the system works internally, including any pre-processing, model architecture or algorithmic stages, and post-processing;

(iii) a step-by-step dry run on at least two representative input images (ideally one good weld and one defective), demonstrating the classification decision and confidence score emitted by the system; and

(iv) the measured inference latency, together with a brief discussion of how the system would integrate with a real conveyor-line controller.

Submissions that do not include the explanation video will be considered incomplete.

8. TIMELINE

The assessment shall be completed and submitted within forty-eight (48) hours from the time of issue. Late submissions will not be considered unless an extension has been agreed in writing in advance.

Note to candidate: *You are encouraged to use any publicly available libraries, frameworks, datasets, or pre-trained models, provided that all such usage is disclosed in your explanation. Originality of approach is valued, but so are engineering judgement, real-time performance, and clarity of communication. If any requirement in this document is unclear, please request clarification before commencing work.*